BT23001CS

BSM/S-24/CS2T006/04

JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

KATOL ROAD, NAGPUR

Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in

(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU & RTMNU

Education to Eternity

॥ज्ञानं सर्वार्थ साधनम्॥

B. Tech Summer / ~~Winter~~ - 2024

(ESE / ~~Make-up~~ / Supplementary) Examination

Branch: Computer Science & Engineering | Sem:- II

Subject: PCC (Data Structure) | Subject Code: CS2T006

Marks: 100 | Date of Exam: 18/06/2024 | Time: 3 Hrs.

Instructions to the students:

1. **Read the question paper carefully (Branch, Semester, Scheme) before attempting the questions.**
2. **All the questions carry marks as indicated.**
3. **Assume suitable data whenever necessary.**
4. **Illustrate your answer whenever necessary with the help of neat sketches.**
5. **Use of Programmable Calculator is prohibited.**
6. **Don't use red pen for writing the answers.**

COs: Students should be able to:

1. Understand basic data structures, their implementation and some of their standard applications.
2. Describe basic data structures as arrays and array operation.
3. Analyze the types of data structures and its operations.
4. Apply linear data structures to various computing problems.
5. Develop programs for Sorting and Searching.

SECTION A

1. **All Questions are compulsory.**
2. **Answer in one/two word/sentence.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Define Data Structure.	L1/CO1	2
Q.2	State the need of data Structure.	L1/CO1	2
Q.3	Give the example of Array.	L2/CO2	2
Q.4	List the characteristics of an Algorithm.	L1/CO2	2
Q.5	Discuss in short Prefix & Postfix expression in Stack.	L2/CO3	2
Q.6	Explain Push & Pop operation of Stack in short.	L2/CO3	2
Q.7	Enlist the application of Stack.	L1/CO3	2
Q.8	Write short note on Priority Queue.	L3/CO4	2
Q.9	Express Enqueue & Dequeue in Queue.	L2/CO4	2
Q.10	Identify the types of Primitive Data Structure.	L1/CO5	2
Q.11	Give any one example of Selection Sort.	L1/CO5	2
Q.12	Rewrite applications of Linear Search.	L2/CO5	2

SECTION B

1. **All Questions are compulsory.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.13	Write short note on Time Complexity and Space Complexity.	L3/CO2	4
Q.14	Analyse various operation in array.	L4/CO5	4
Q.15	Rewrite the advantages & disadvantages of Stack.	L5/CO3	4
Q.16	Explain the types of Queue with example.	L2/CO4	4
Q.17	Illustrate the conversion of Infix into Postfix expression	L3/CO3	4
Q.18	Examine the advantages & disadvantages of Binary Search.	L4/CO5	4

SECTION C

1. **Solve any FOUR Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.19	Rewrite various types of data structure operations in detail.	L5/CO5	7
Q.20	Write types of Arrays with an example.	L5/CO5	7
Q.21	Illustrate in detail the types of Stack Operations.	L4/CO3	7
Q.22	Describe the issues of Queue in detail.	L6/CO4	7
Q.23	Explain Binary search with example.	L5/CO5	7
Q.24	Distinguish between Searching and Sorting Algorithm.	L4/CO5	7

O(N) O(N²)

SECTION D

1. **Solve any TWO Questions.**

Q. No.	Questions	Level/CO Mapped	Marks
Q.25	Illustrate Linear search with example also write their advantages & disadvantages.	L4/CO5	12
Q.26	Describe various applications of Queue in data structure in detail.	L6/CO4	12
Q.27	Explain in detail Sorting algorithm with their types and example.	L5/CO5	12